

# Mohamed Ashfaq

Computer Science Undergraduate Student

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)



## PROFILE

Dedicated and project-driven Computer Science Undergraduate specializing in Full-Stack Development with a Back-End focus. Proven ability to translate complex requirements into scalable and functional code, as demonstrated through academic projects utilizing robust server-side technologies like Java Spring Boot, paired with modern frameworks such as Next.js and React. Proficient in persistent data management using MySQL and PostgreSQL. Seeking an entry-level software engineering position where technical proficiency and a commitment to clean, efficient code can drive immediate value.

## STRENGTHS AND EXPERTISE

**Languages:** Java, Python, JavaScript, TypeScript, C++, HTML5, CSS3

**Back-End/Databases:** Spring Boot, MySQL, PostgreSQL

**Front-End Frameworks:** React, Next.js

**Core Concepts:** OOP (Object-Oriented Programming), DSA (Data Structures & Algorithms)

**Soft Skills:** Team Management, Problem Solving, Team Work, Adaptability

**Other Skills:** Git / GitHub, Postman, Figma, Microsoft Office (Word, Excel, PowerPoint)

## EDUCATION

**Informatics Institute of Technology Affiliated to University of Westminster**  
**BSc (Hons) in Computer Science**

Sep 2024 - present

Module Marks - Year 01 / Level 04 | Average: 76.8

- Software Development I (Python) : 78
- Software Development II (java) : 83
- Web Design & Development : 65
- Computer System Fundamentals : 97
- Mathematics for Computing : 88
- Trends in Computer Science : 50

Module Marks - Year 02 / Level 05

- Object Oriented Programming: 78
- Database Systems: 87
- Software Development Group Project: pending
- Machine Learning and Data Mining: pending
- Algorithms: Theory Design and Implementation: pending
- Client Server Architecture: pending

**Hindu College, Rathmalana**

Jan 2010- Dec 2023

**G.C.E. Advanced Level (Physical Science)**

- Achieved strong passes in the Physical Science stream, including Combined Mathematics and Physics (2 C, 1S).

**G.C.E. Ordinary Level**

- Demonstrated foundational academic strength with an A Grade in Mathematics and strong passes in key subjects (1 A, 7 C, 1 S total).

## PROJECTS

**Airea - Smart Respiratory Monitor for Early Lung Cancer Detection (Ongoing) | SDGP Team Project|**

**Tech Stack:** C++ (ESP32), Flutter, Python (Backend), Firebase (FCM, Firestore)

The project involved engineering the C++ firmware for the ESP32 multisensory wearable device to continuously capture and stream patient vital signs, including SpO<sub>2</sub> and Heart Rate, alongside autonomously detecting patient falls. The system's back-end was designed using Python to perform real-time analysis of long-term cough patterns and trends to inform early lung cancer detection models. This data is delivered to caregivers and patients via a Flutter mobile application, which displays real-time data and issues instant, actionable risk alerts using Firebase Cloud Messaging (FCM), with all patient data securely and persistently stored in Supabase.

**Eco Pulse – Sustainable Development Web Platform | Team Lead |**

**Tech Stack:** HTML5, CSS3, Vanilla JavaScript, Figma

Directed a team of 4 in building a responsive web platform advocating for UN SDG 13 (Climate Action). Architected the project entirely from scratch without frameworks to demonstrate deep mastery of core web technologies and W3C validation standards. Implemented a global CSS design system for visual consistency and developed interactive components, including real-time form validation and dynamic grid layouts.

### **Traffic Data Analyst – Python-Based Visualization Tool | Solo Project |**

**Tech Stack:** Python, Tkinter, Pandas, Data Visualization

Developed a standalone Python Tkinter application designed for efficient handling and visualization of traffic junction data. The program retrieves date-specific records, utilizes fundamental Python data structures and loops for reliable processing, and displays the summarized data in the console. The core feature is the in-app visualization of the data using a histogram, demonstrating proficiency in data manipulation and GUI development.

### **Black Olives – Premium Apple Retail Platform | Solo Project |**

**Tech Stack:** Next.js

Engineered a high-performance e-commerce application tailored for a premium Apple reseller. Designed a minimalist, dark-themed UI with dynamic product navigation and CSS Modules to ensure a seamless luxury shopping experience across devices. Leveraged Next.js server-side rendering (SSR) to optimize SEO visibility and load speeds, delivering a responsive single-page application (SPA) feel.

### **SpendWise – Distributed Full-Stack Expense Tracker | Solo Project |**

**Tech Stack:** Java (Spring Boot), React.js, PostgreSQL, REST APIs

Developed and deployed a scalable personal finance application using a decoupled 3-tier architecture. Built a robust RESTful API with Spring Boot to handle complex business logic and expense data processing, integrated with a React frontend for real-time interaction. Implemented a distributed cloud deployment strategy, hosting the frontend on Netlify, backend on Railway, and the persistence layer on Supabase.

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## **ACHIEVEMENTS**

- Participated in SLIIT MATHFEST 2023
- Successfully participated in the final stage of the CodeRally 6.0 and Xtreme 19.0 Hackathons, focusing on rapid development and collaborative problem-solving under pressure.
- Won multiple certificates at various annual prize givings.

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## **EXTRA CURRICULAR ACTIVITIES**

### **INFORMATICS INSTITUTE OF TECHNOLOGY**

Member | IEEE (2024 - present)

- CodeRally 6.0 | Program
- Xtreme 19.0 | Program

### **Hindu College, Rathmalana**

- Prefect(2019-2022)
- Event coordinator

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## **CERTIFICATIONS**

- Typescript for JavaScript Developers (LinkedIn Learning, 2025)
- Critical Thinking and Problem Solving (LinkedIn Learning, 2024)
- React Essential Training (LinkedIn Learning, 2025)
- Python GUI Development with Tkinter (LinkedIn Learning, 2024)
- Learning Python (LinkedIn Learning, 2024)

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## **REFERENCE**

### **Aakash Nambiyarooran**

Associate Project Manager/

hSenid Mobile Solution. (pvt) Ltd

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### **Chalindu Yohan A. Fernando**

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